

SEC P vs NP log 2026-04-28

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-28 01:39 UTC

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1 cycles recorded

Block-Composition Sub-Additivity of Coarse Depth κ via the Roe-Trace Pairing

- **Verdict:** INCONCLUSIVE
- **Bridge:** Coarse Geometric Karchmer-Wigderson (CG-KW); Roe-style coarse cohomology / coarse geometry of metric spaces (geometric group theory + index theory branch of analysis) $\times$ Boolean formula depth lower bounds for KRW-style block compositions $f \circ g$, the central complexity-theoretic object behind $P \not\subseteq NC^1$; equivalently the depth complexity $D(f \circ g)$ of the composed function in the Karchmer-Raz-Wigderson program
- **Recorded:** 2026-04-28 01:39 UTC
- **Entry ID:** 8047f7f8ee1

Statement

For all boolean functions $f: \{0,1\}^n \rightarrow \{0,1\}$ and $g: \{0,1\}^m \rightarrow \{0,1\}$ for which $HX^1(X_f)$ and $HX^1(X_g)$ admit nontrivial classes detected by Roe-controlled operators, the coarse depth κ defined via the trace pairing on the KW-metric spaces obeys the sub-additive bound $\kappa(f \circ g) \geq \kappa(f) + \kappa(g) - C$, where C is a universal constant independent of n, m , and where $(X_{\{f \circ g\}}, d_{\{f \circ g\}})$ is built by the coarse product operation. In particular, witness cocycles $c_f \boxtimes c_g \in HX^1(X_{\{f \circ g\}})$ and tensor Roe operators $T_f \otimes T_g \in C_u^*[X_{\{f \circ g\}}]$ realize $\log|(c_f \boxtimes c_g, T_f \otimes T_g)| / \log(\text{propagation}(T_f \otimes T_g)) \geq \kappa(f) + \kappa(g) - C$.

Rationale

This sub-conjecture lifts axiom A2 (Composition sub-additivity for asdim) from the asymptotic-dimension layer to the strictly stronger κ layer, and chains through A1 ($\kappa \leq \text{depth}$) to recover the KRW conjecture $D(f \circ g) \geq D(f) + D(g) - O(1)$. The mechanism is the multiplicativity of the Roe-index trace pairing on tensor products: $\langle c_f \boxtimes c_g, T_f \otimes T_g \rangle = \langle c_f, T_f \rangle \cdot \langle c_g, T_g \rangle$ (Connes-style product), while the propagation of $T_f \otimes T_g$ equals $\text{propagation}(T_f) \cdot \text{propagation}(T_g)$ under the depth-additive metric of coarse product. Taking logs yields the additive bound. Should this hold even on small toy compositions, it provides direct empirical support for the KRW direct-sum theorem at the κ level. It does not contradict A3 (anti-natural-proofs) because random f has $\kappa = O(1)$, so additivity holds trivially. It is consistent with A4 (anti-relativization) since coarse product does not introduce oracle gates. A5 (Roe-index rigidity) underpins the stability of the tensor pairing under coarse equivalence.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6db712c6.py", line 244, in <module>
    result = run_trial(seed)
    ^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6db712c6.py", line 186, in run_trial
    kappa_f, kappa_g, kappa_fg = run_composition(f, g, k_f, k_g)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6db712c6.py", line 143, in run_composit
    X_f, d_f = build_X_d(f, k_f)
    ^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6db712c6.py", line 43, in build_X_d
    X = [tuple(i) for i in range(n)]
    ^^^^^^^
```

TypeError: 'int' object is not iterable

Judge reasoning

The test crashed with a `TypeError` ('int' object is not iterable) at `build_X_d` before any κ values were computed, so no data exists to evaluate the pre-registered support criterion. | next: Fix `build_X_d` to enumerate vertices correctly (e.g., `X = [tuple(b) for b in itertools.product([0,1], repeat=n)]`) and rerun the 25-pair $\times$ 5-seed protocol.